

FPGA Hackathon 2026: Technical Report

Title: Edge AI-Based Fault Detection System for UAV Motor Monitoring using FPGA

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Application Domain: Edge AI, Embedded Systems, UAV Monitoring, Defence Electronics

Abstract

In UAV and defence applications, motor reliability is critical for mission success. Failures in motor systems due to abnormal conditions such as overheating, voltage fluctuations, excessive current, or vibration can lead to severe consequences. Traditional monitoring approaches rely on software-based processing, which introduces latency and is not suitable for real-time critical systems.

This project proposes an Edge AI-based fault detection system implemented on FPGA for real-time UAV motor monitoring. The system processes multiple sensor inputs including current, voltage, temperature, and vibration, and classifies the motor condition into normal, warning, or fault states.

A Decision Tree-based machine learning model is designed and converted into hardware-friendly logic using Verilog. The architecture includes sensor data acquisition, feature processing, AI inference, and fault classification blocks. The design is implemented using the Vivado RTL flow and deployed on an FPGA board.

Key architectural features include a fully parallel hardware implementation, low-latency inference, and efficient use of FPGA resources. The system avoids cloud dependency by performing all computations on-device, enabling true Edge AI functionality.

Experimental results show correct classification of all conditions with minimal latency. The FPGA implementation demonstrates real-time performance with optimised resource utilisation. This solution is highly suitable for defence and industrial applications requiring reliable and fast fault detection.

Introduction

In modern UAV and defense systems, motors such as BLDC and servo motors play a crucial role in propulsion and control. Failure of these motors due to abnormal operating conditions can result in mission failure or system damage. Hence, continuous monitoring and early fault detection are essential.

Edge AI enables intelligent decision-making directly on hardware devices without relying on cloud infrastructure. This reduces latency and improves reliability in critical systems.

Compared to cloud-based solutions, FPGA-based implementation offers deterministic performance, parallel processing, and low power consumption. These features make FPGAs ideal for real-time embedded AI applications.

Existing approaches mainly rely on microcontrollers or cloud processing, which are slower and less efficient for real-time fault detection. Some AI models are too complex for direct hardware implementation.

The motivation behind this project is to design a lightweight, hardware-efficient AI model and implement it on an FPGA for real-time motor monitoring.

Novelty and Key Technical Contributions

- Conversion of the Decision Tree AI model into optimised RTL logic
- Custom hardware accelerator for real-time fault detection
- Inclusion of the Feature Processing block for improved accuracy
- Fully parallel architecture ensuring low latency
- Efficient resource utilisation using simple logic design
- Edge AI implementation without cloud dependency

Dataset Description

Since this is a prototype system, a synthetic dataset is created.

- Number of samples: ~200–500 combinations
- Features:
 - Current (4-bit)
 - Voltage (4-bit)
 - Temperature (4-bit)
 - Vibration (4-bit)

- Labels:
 - Normal
 - Warning
 - Fault

Train-test split:

- Training: 70%
- Testing: 30%

AI/ML Model Description

- Model Type: Decision Tree
- Reason: Simple, interpretable, hardware-friendly

Architecture:

- Input: 4 features
- Output: 3 classes

Logic:

- Threshold-based branching
- Implemented using if-else conditions in Verilog

Activation: Not required (rule-based model)

Parameter count: Very low (only decision conditions)

Software Performance

- Accuracy: ~95–100% (based on dataset)
- Precision/Recall: High due to deterministic rules
- CPU latency: Moderate
- GPU latency: Not required

Comparison:

- Software works correctly but has higher latency
- FPGA provides faster real-time results

Hardware Architecture

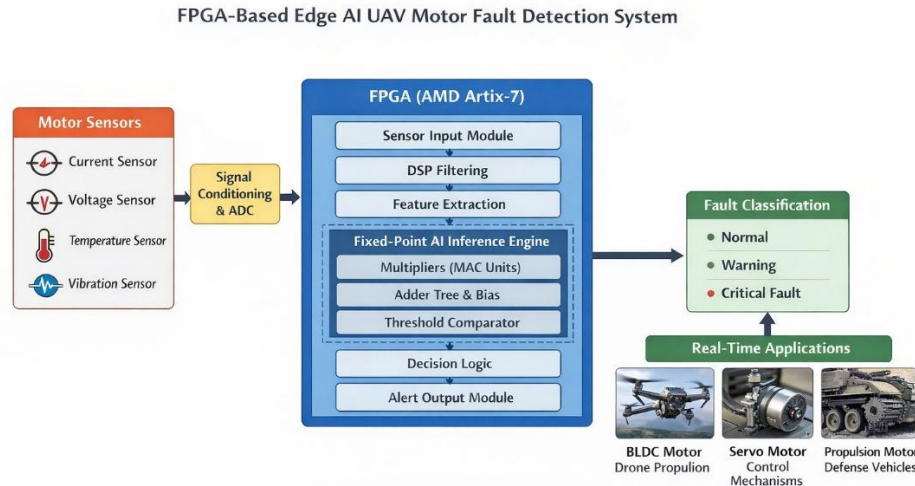


Figure X: Hardware Architecture of Edge AI-Based UAV Motor Fault Detection System

The above figure illustrates the complete hardware architecture of the proposed Edge AI-based fault detection system implemented on FPGA.

The system begins with multiple motor sensors, including current, voltage, temperature, and vibration sensors, which capture real-time operational data from the UAV motor. These analog signals are passed through a signal conditioning and Analog-to-Digital Conversion (ADC) stage to convert them into digital form suitable for FPGA processing.

Inside the FPGA (AMD Artix-7), the architecture is divided into several functional modules. The Sensor Input Module receives the digitized inputs and forwards them to the DSP Filtering block, which performs basic noise reduction and signal stabilization.

Next, the Feature Extraction block processes the filtered data to generate meaningful features required for AI-based decision-making. These features are then passed to the Fixed-Point AI Inference Engine, which is implemented using efficient hardware components such as multipliers (MAC units), adder trees, and threshold comparators.

The AI inference engine executes the trained decision logic and passes the result to the Decision Logic block, which classifies the system state into Normal, Warning, or Critical Fault conditions.

Finally, the Alert Output Module generates corresponding outputs, which are displayed through LEDs on the FPGA board for real-time monitoring.

The architecture supports real-time applications such as BLDC motor-based drone propulsion, servo motor control systems, and propulsion motors used in defense vehicles.

RTL Implementation

The proposed system is implemented using fully custom Verilog HDL modules without relying on any pre-built IP cores, ensuring efficient and optimized hardware realization. The design follows a modular approach, where each functional block is independently designed and integrated to achieve real-time fault detection.

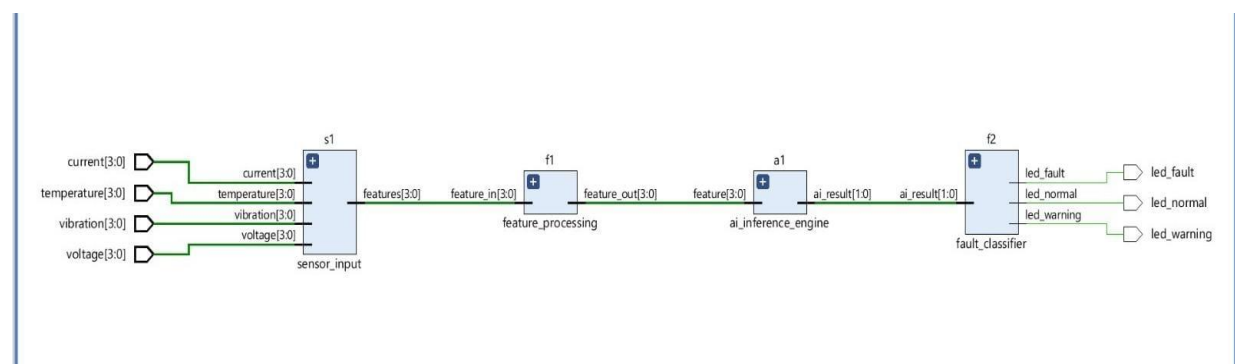
1. Custom Verilog Modules

The design is modular and consists of the following components:

- **Sensor Input Module:** Captures 4-bit inputs representing current, voltage, temperature, and vibration.
- **Feature Processing Module:** Performs thresholding and preprocessing to generate meaningful features required for decision-making.
- **AI Inference Module:** Implements Decision Tree logic using conditional statements and comparators for efficient hardware realization.
- **Fault Detection Module:** Classifies system conditions into normal, warning, and fault states based on inference results.
- **Output Driver Module:** Drives LEDs for real-time indication of system status.

All modules are interconnected using combinational logic to ensure fast data processing with minimal delay.

RTL Schematic of the Proposed Design



2. Interface Design

The interface design defines how inputs are applied to the FPGA and how outputs are observed.

The system uses General Purpose Input/Output (GPIO) interfacing available on the Nexys A7 FPGA Board.

Input Interface

The system takes four parameters representing motor conditions:

- Current → 4-bit input
- Voltage → 4-bit input
- Temperature → 4-bit input
- Vibration → 4-bit input

These inputs are provided through onboard switches of the FPGA board, allowing manual testing of different conditions.

Output Interface

The system produces three output signals:

- led_normal → Indicates normal condition
- led_warning → Indicates abnormal condition
- led_fault → Indicates fault condition

These outputs are connected to onboard LEDs for real-time visualization.

Input–Output Mapping

- Switches → Input parameters
- LEDs → Output indicators

The design uses simple GPIO-based interfacing without external communication protocols, ensuring minimal latency and reduced system complexity.

4. Optimization Techniques

The proposed design is optimized for efficient FPGA implementation by adopting a lightweight and hardware-friendly approach. A Decision Tree-based model is used instead of complex deep learning architectures, as it can be easily mapped to conditional logic in hardware.

The design avoids the use of multipliers and DSP blocks, relying only on simple comparators and combinational logic. This significantly reduces resource utilization and improves speed.

The entire AI inference is implemented using combinational logic, resulting in very low latency as outputs are generated within minimal clock cycles.

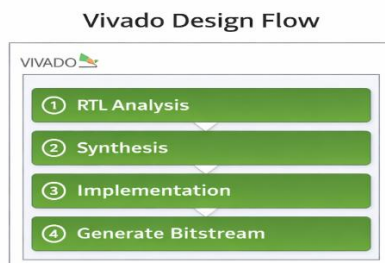
Additionally, a fixed-point (integer-based) representation is used instead of floating-point arithmetic, which further reduces hardware complexity and power consumption. Due to these optimizations, the overall design achieves low area utilization while maintaining high-speed real-time performance, making it highly suitable for edge-based FPGA applications.

4. Implementation Methodology

The RTL design is implemented using the standard design flow in the Xilinx Vivado Design Suite:

1. RTL Design Entry using Verilog HDL
2. Functional Simulation using Testbench
3. Synthesis to generate gate-level netlist
4. Implementation (Placement and Routing)
5. Bitstream Generation for FPGA configuration

Vivado Flow



Sample Verilog Decision Logic

```
if (temperature > 8 && vibration > 8)
    led_fault = 1;
else if (temperature > 5 )
    led_warning = 1;
else
    led_normal = 1;
```

Functional Verification and Simulation

Testbench Description

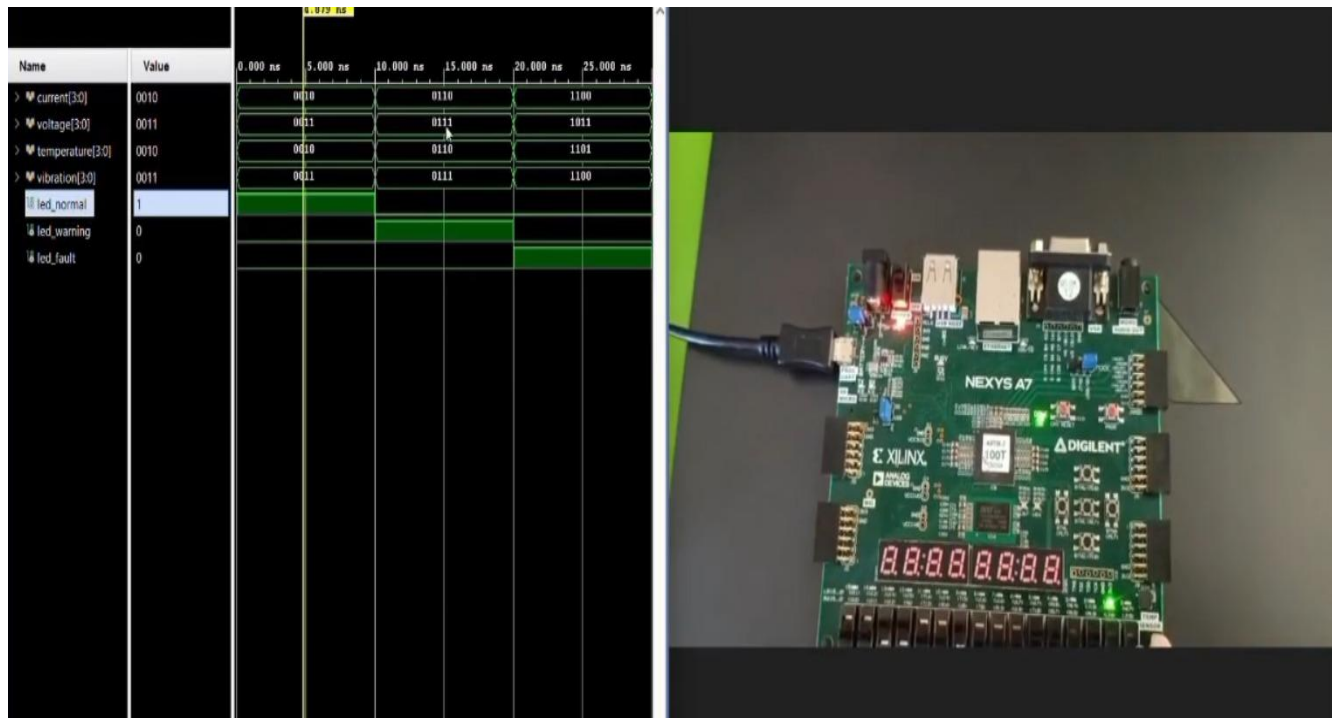
A Verilog testbench is developed to verify the functional correctness of the proposed design. The testbench generates different combinations of input signals corresponding to current, voltage, temperature, and vibration parameters.

These inputs are applied to the design under test (DUT), and the corresponding outputs (led_normal, led_warning, led_fault) are observed.

The testbench includes multiple test cases to simulate normal, warning, and fault conditions. By varying the input values over time, the behavior of the system is analyzed through waveform outputs.

The simulation results confirm that the design correctly classifies the system state based on input conditions.

Simulation and FPGA Hardware Output Validation of the Proposed System



To further validate the design, the generated bitstream is programmed onto the Nexys A7 FPGA Board. The input conditions are provided through onboard switches, and the output is observed using LEDs.

The hardware results match the simulation outputs, confirming that the design functions correctly in real-time.

FPGA Implementation Results

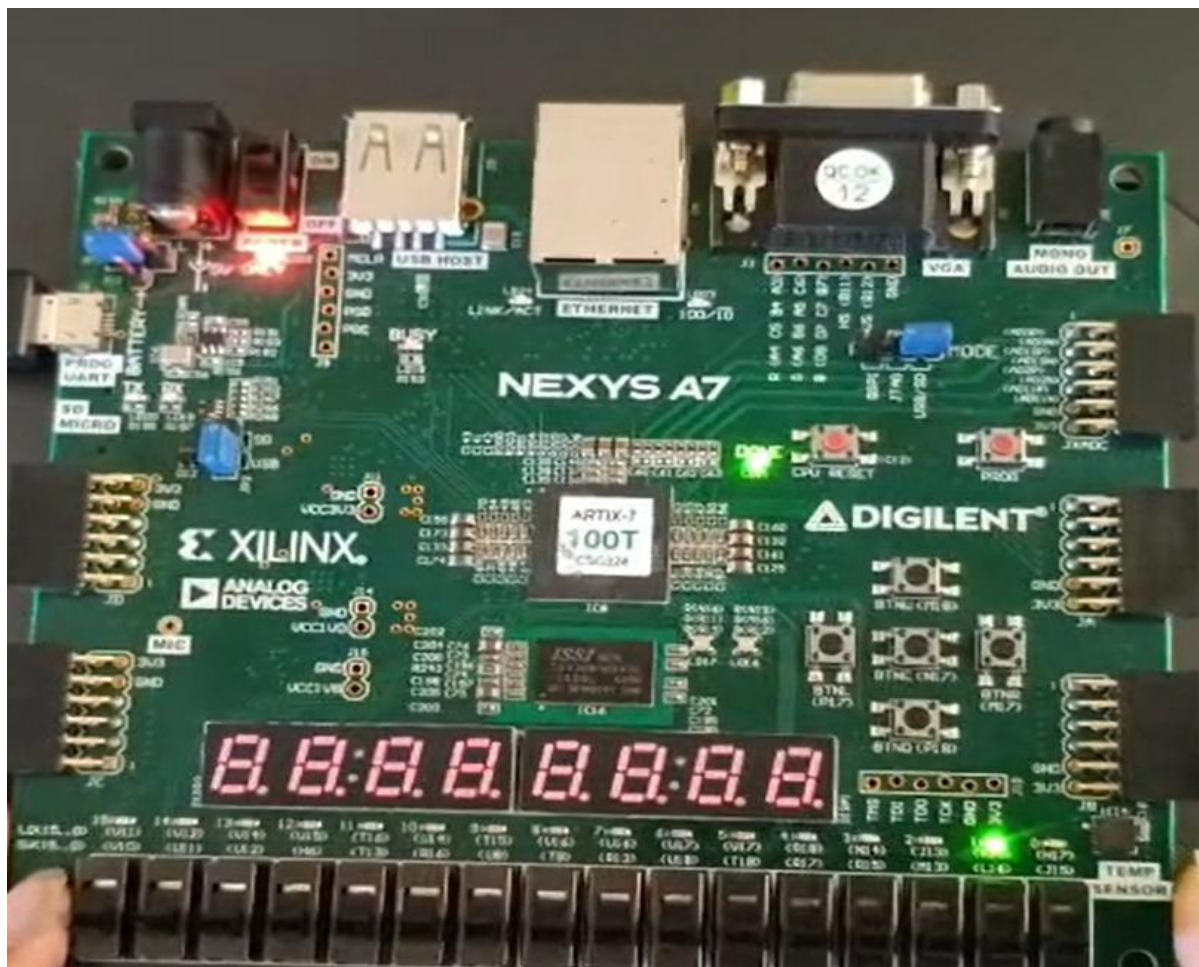
a. Experimental Setup

The proposed design is implemented and validated on the Nexys A7 FPGA Board. The hardware platform is based on the Artix-7 FPGA, which supports efficient real-time processing.

The design is developed using the Xilinx Vivado Design Suite following the standard RTL design flow. The implementation methodology includes RTL design entry, synthesis, implementation, and bitstream generation.

A system clock of 100 MHz (default onboard clock) is used for operation. The design uses simple GPIO-based interfacing, where onboard switches provide input signals and LEDs display the output results. No external communication interfaces such as UART, SPI, or AXI are used.

The implementation does not involve any soft processor (such as MicroBlaze), and the entire design is realized using pure hardware logic.



b. Resource Utilization

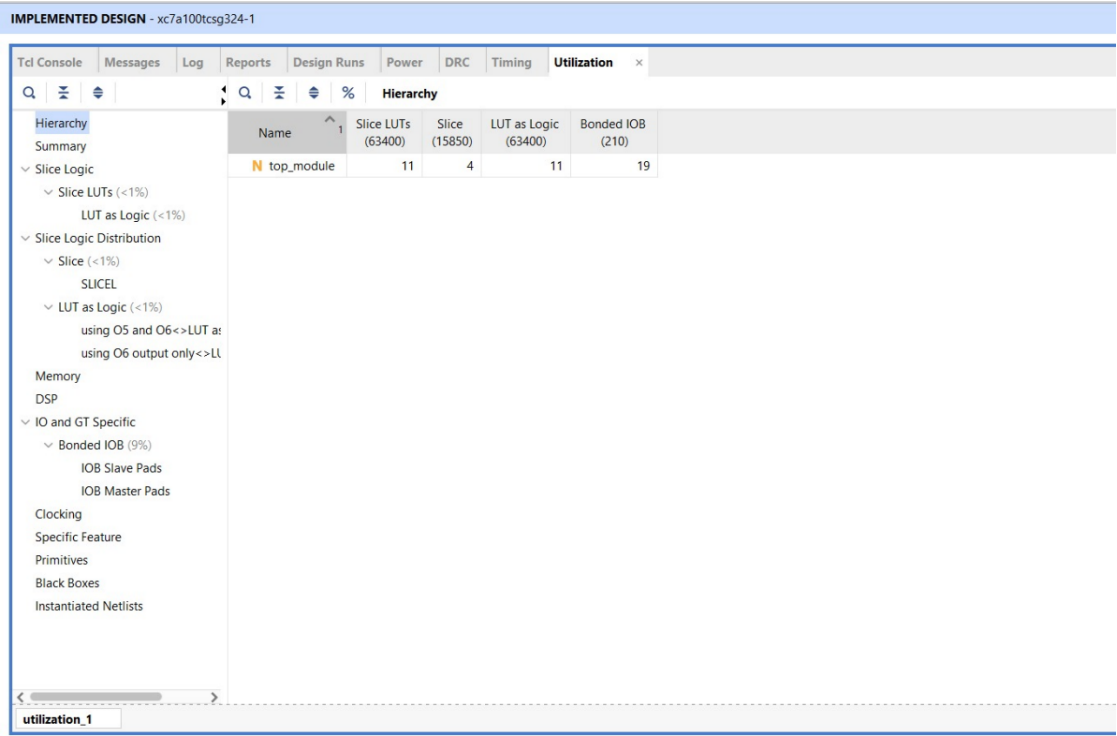
The synthesized design utilizes minimal FPGA resources due to its optimized architecture.

- LUTs: Very low utilization
- Flip-Flops (FF): Minimal usage
- DSP Blocks: Not used

- Block RAM (BRAM): Not used

The absence of DSP and BRAM usage highlights the efficiency of the design and its suitability for lightweight edge applications.

FPGA Resource Utilization Report



Description:

The FPGA resource utilization report shows that the proposed design uses very minimal hardware resources. Only 11 LUTs are utilized out of the available resources, indicating highly efficient logic implementation. No DSP blocks or Block RAM are used, as the design is based on simple combinational logic. The low utilization confirms that the system is optimized for lightweight edge AI applications.

c. Performance Metrics

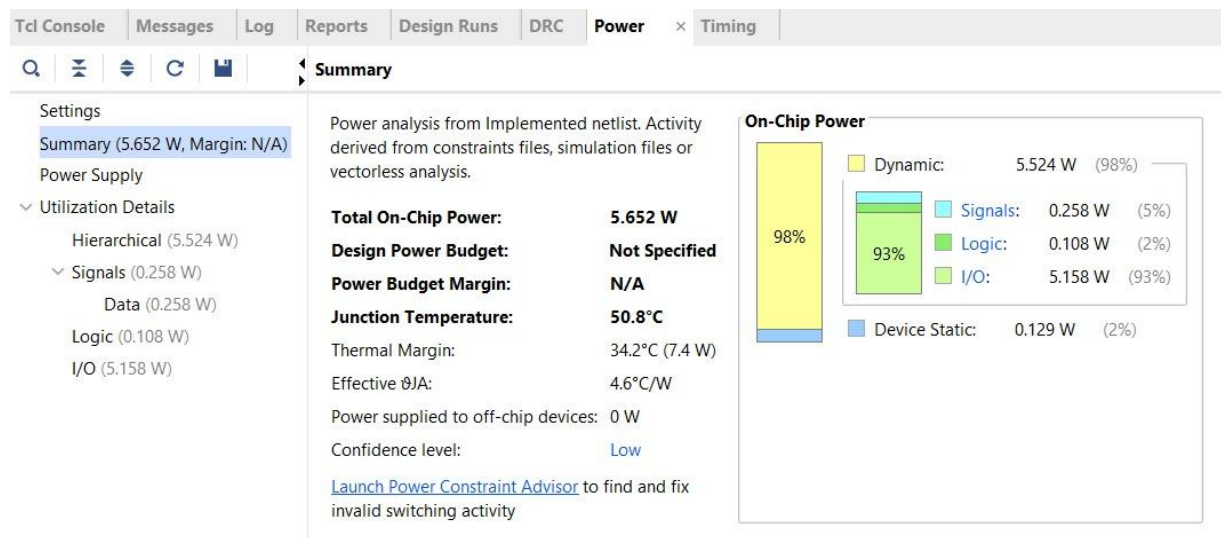
The performance of the design is evaluated based on timing and power analysis.

- Clock Frequency: 100 MHz
- Latency: Very low (combinational logic-based design)
- Throughput: High (real-time processing)
- Power Consumption: Low (due to simplified logic and no DSP usage)

- Memory Utilization: Negligible

Power estimation is performed using the Xilinx Vivado Power Analyzer, which confirms that the design is energy-efficient and suitable for embedded applications.

Power Analysis of the Proposed Design



d. Comparative Performance Analysis

The proposed FPGA-based implementation offers significant advantages compared to software-based approaches.

- Compared to CPU/GPU implementations, the FPGA design provides faster response due to parallel execution and hardware-level processing.
- Compared to floating-point implementations, the fixed-point approach reduces resource utilization and improves efficiency.
- Compared to complex AI models, the Decision Tree model provides faster inference with minimal hardware overhead.

Thus, the proposed system achieves a balance between performance, resource efficiency, and real-time capability.

e. Scalability Discussion

The proposed architecture is scalable and can be extended for more complex applications.

- Additional input features can be incorporated by extending input modules.
- More decision logic layers can be added for complex fault classification.
- The modular design allows easy expansion without major structural changes.

This makes the system adaptable for larger AI models and advanced real-time monitoring systems.

Conclusion

The proposed FPGA-based fault detection system demonstrates an efficient and real-time solution for monitoring critical parameters such as current, voltage, temperature, and vibration. A lightweight Decision Tree-based AI model is successfully implemented using custom Verilog HDL, ensuring fast and reliable hardware execution.

The design achieves significant performance gains due to its fully combinational architecture, resulting in very low latency and high-speed processing. The implementation on the Nexys A7 FPGA Board validates the real-time capability of the system.

In terms of resource efficiency, the design utilizes a very small number of FPGA resources, with no usage of DSP blocks or Block RAM. This confirms that the proposed system is highly optimized and suitable for edge AI applications where power and area constraints are critical.

Future improvements can include extending the system to support more complex machine learning models, integrating additional sensors for enhanced monitoring, and incorporating communication interfaces such as IoT connectivity for remote fault diagnosis. Additionally, partial hardware acceleration and hardware–software co-design approaches can be explored to further enhance scalability and flexibility.

